YELP COMPLAINT TAGGER · DEMO

Reads bad restaurant reviews and tells you why.

huggingface.co/nrr26/yelp-complaint-tagger

A learning project, shared to learn in public

HOW I BUILT IT

Five steps, one day, free tools.

01
Rules
8 complaint types + clear rules

02
Labels
Free AI labelled 3,500 reviews; ~18/20 right by hand

03
Baseline
Keyword model: F1 0.58

04
Training
Small DistilBERT model: F1 0.71

05
Publish
Honest model card on Hugging Face

Keyword baseline

0.58

→

My model

0.71

Average F1 score on 354 held-out reviews.
Better on all 8 complaint types.

TOOLS AND TIME

Everything was free. It took about 6 hours.

Google Colab
Free T4 GPU for labelling and training

Hugging Face Hub
Private dataset + public model hosting

Qwen3-4B-Instruct
Free open LLM that labelled 3,500 reviews

DistilBERT
66M-parameter model I fine-tuned

Python libraries
transformers, datasets, scikit-learn, pandas

Claude
AI coach for each step; git for versions

Label guide + setup	45 min
Data + pilot labelling	1.5 h
Full labelling run	40 min
Baseline + training	1 h
Checks + model card	1 h
Publish	15 min

Build your own: 1–2 weeks
Collect your own reviews, hand-check a few hundred labels, then train. Training itself takes minutes.

HOW IT COULD IMPROVE

Five upgrades for version 2.

01

A bigger human-checked test set

Score against people, not only the AI labeller

02

Add happy and neutral reviews

So it can say “no complaint”

03

More rare examples

Billing and wrong orders need more data

04

Show the evidence

Highlight the sentence behind each tag

05

A weekly dashboard

Run it on every new review automatically

Known limits today: complaints only (not 5-star reviews), trips on negation, and copies some habits of its AI labeller.

WHERE THIS HELPS

Six practical use cases.

Owners and managers

Weekly complaint breakdown: fix the right thing first

Groups and franchises

Compare locations; send help where wait times spike

Delivery platforms

Route each complaint to the right team

Reputation agencies

Hundreds of reviews into a one-page monthly report

Restaurant software

Review insights inside POS or reservation tools

Food-safety teams

Surface hygiene complaints fast for follow-up

Beyond restaurants: hotels, clinics, retail, winery tasting rooms. Change the labels, get a new model.

Commercial use needs a version trained on your own reviews; this one is non-commercial.